Follow the Workers, Not the Bids: What Employment Records Reveal about Procurement Cartels*

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Abstract

Brazil's antitrust authority convicted 35 firms across seven procurement cartels in São Paulo between 2009 and 2019. We link these firms to 2.6 million competitive auctions on the state's electronic platform and to the universe of matched employer–employee records, assembling the first detection benchmark to combine auction data, labor microdata, and formal antitrust rulings. Six of the seven cartels have auctions in our close-margin evaluation sample.

On roughly 2,300 close-margin auctions involving at least one convicted firm, canonical variance screens yield AUCs of 0.46–0.56 in the aggregate—but this null conceals a sharp split by coordination technology. On bid-rigging cartels, classical screens achieve AUCs of 0.65; on the hub-and-spoke cartel that dominates the sample (85%), they are at chance. The close-margin RD of Kawai et al. (2023) produces a null on both types. Worker-flow screens—counting employees who have worked at both the winning and runner-up firms—appear far more powerful, with an AUC of 0.75.

That headline result does not survive temporal conditioning. When we restrict worker flows to those observed *before* each auction, the AUC falls to 0.50—indistinguishable from chance. A temporal decomposition reveals where the signal actually lives: not in pre-existing personnel connections that might reflect coordination, but in post-conduct labor reallocation. After cartel conduct periods end, workers flow from convicted firms to their rivals at a ratio that rises from 1.4:1 to 4:1 over the panel, consistent with enforcement-driven displacement rather than collusive coordination. The post-conduct

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component alone produces an AUC of 0.77; the pre-conduct component produces exactly 0.50.

The finding carries two implications. First, detection benchmarks built from panel data are vulnerable to look-ahead bias: network screens that use the full employment history mechanically absorb enforcement aftermaths and overstate prospective detection power. This caveat applies to any screen—worker-flow, ownership, or board-interlock—evaluated against conviction-anchored ground truth without temporal conditioning. Second, the temporal pattern is itself informative: cartel enforcement in thin labor markets triggers a measurable reallocation of human capital from convicted firms to competitors, a real effect that matched employer–employee records are uniquely positioned to document.

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1. Introduction

Can the canonical tools for detecting procurement cartels distinguish collusion from noise when they leave the thick-market, high-enforcement environments in which they were built? And when a new screen appears to succeed where they fail, is the signal genuine—or an artifact of how the benchmark was constructed? We address both questions by linking seven cartels convicted by Brazil's antitrust authority to 2.6 million competitive auctions and the full universe of formal employment relationships in São Paulo. The answer to the first question is discouraging: classical variance screens and a close-margin regression discontinuity yield AUCs of 0.46–0.56. The answer to the second is a warning: worker-flow screens constructed from the same data appear to achieve an AUC of 0.75, but temporal conditioning reveals that the entire signal comes from labor reallocation *after* cartel conduct ends—not from pre-existing personnel connections that might reflect coordination. The traces that employment records capture are not those of collusion operating, but of collusion dissolving.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly. Classical screens exploit the compression that bid-rotation agreements impose on the bid distribution: prearranged prices cluster around the designated winner, yielding low dispersion, few active bidders, and predictable rotation (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). The complementary approach of Kawai et al. (2023) focuses on the winning–losing threshold: under genuine competition, firms that barely win and barely lose should look alike, and any sharp discontinuity at the cutoff identifies a non-competitive wedge. Both families of tests have been validated primarily against cartels in high-income economies with deep bidder pools and strong institutional capacity for enforcement; their track record in thinner settings rests on a small and growing empirical base.

Network-based methods have pushed the frontier in two directions. Within the procurement data itself, co-bidding networks flag candidate cartels through group-level cohesion and exclusivity (Wachs and Kertész, 2019; Signor et al., 2020; Huber and Imhof, 2019; Clark et al., 2018). Outside the bid record, a recent wave exploits external network layers: Asai and Charoenwong (2026) show that common-ownership links predict bid rigging in Singapore, and Villamil et al. (2024) build multiplex ownership–co-bidding networks for Swedish procurement. What none of these studies does—and what our

benchmark uniquely enables—is test whether the detection signal from a network layer survives temporal conditioning: restricting the network measure to information available *before* the auction being screened. This distinction matters because network panels built from administrative registries (employment, ownership, board membership) mechanically include post-event observations, and any correlation between the network measure and the ground-truth label may reflect the consequences of detection and enforcement rather than the coordination the screen is meant to identify.

Our primary evaluation sample is a *cartel-active* benchmark of roughly 2,300 close-margin pregão auctions in which at least one convicted firm bid during the active period of its cartel (details of sample construction in Section 5). On that benchmark, classical screens disappoint. Bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions (Cohen's d between -0.07 and -0.24), and Welch t-statistics are significant thanks to the enormous control sample—but the AUCs top out at 0.56, far short of what a triage tool would require. The close-margin RD produces a null. In thin markets—where the median number of bidders falls below five—classical screens actually fire in the *wrong* direction (AUC 0.40–0.45), consistent with phantom-bidding mechanisms that inflate rather than compress the bid distribution.

Worker-flow screens initially appear to resolve this failure. Counting the employees who have worked at both the winning and the runner-up firm over the 2009–2017 RAIS panel yields an AUC of 0.75. But this headline rests on the full nine-year employment panel, which includes worker movements that occurred years *after* the auction being screened. When we condition on temporal precedence—counting only workers who had appeared at both firms *before* the auction year—the AUC falls to 0.52, indistinguishable from chance. A three-year rolling window produces an essentially identical result (0.54). The prospective detection power of worker-flow screens is, on this benchmark, zero.

The temporal decomposition reveals where the signal actually lives. We classify each shared worker by when the inter-firm connection was first observable, relative to the cartel's conduct period. Pre-conduct worker flows produce an AUC of exactly 0.50. During-conduct flows produce 0.62. Post-conduct flows—worker movements that occurred *after* the cartel period ended—produce 0.77, accounting for nearly all of the headline result. The flows are directional: workers move predominantly *from* convicted cartel firms *to* their market rivals, at a ratio that rises from 1.4:1 in 2010 to 4:1 by 2016–2017. This pattern is consistent with enforcement-driven labor reallocation: as investigations proceed and conduct periods end, cartel firms shed employees who are absorbed by competitors

in the same thin labor markets.

The paper makes four contributions. The first is a **positive detection result**. Although classical screens fail in the aggregate, disaggregating by coordination technology reveals that they detect bid rigging at operationally useful levels (AUC ≈ 0.65). The aggregate null is a composition effect: the hub-and-spoke cartel that dominates the sample (85%) leaves no trace in bid patterns and dilutes the bid-rigging signal. Matching the screen to the coordination technology restores detection power for the cartel types that classical screens are designed to catch.

The second is **methodological**. We document that detection benchmarks built from administrative panel data are vulnerable to look-ahead bias. When a network screen (worker-flow, ownership, or board-interlock) is evaluated against conviction-anchored ground truth using the full panel, post-enforcement labor reallocation inflates the apparent AUC. Temporal conditioning—restricting the network measure to pre-auction information—is necessary to distinguish prospective detection power from retrospective correlation. This caveat applies broadly: any study that evaluates a panel-based network screen against ground truth without conditioning on temporal precedence risks overstating detection performance.

The third contribution is **substantive**. The temporal pattern we document—post-conduct worker flows from cartel firms to rivals, intensifying over time and strongly directional—is evidence that enforcement reshapes the labor-market structure of thin product markets. Szerman (2023) showed that debarred procurement firms lose nearly half their employees; we show where those employees go: to competitors operating in the same specialized product markets. The matched employer–employee record thus captures not the coordination channel of collusion but its labor-market aftermath.

The fourth contribution is **empirical**. We deliver the first stress test of canonical detection frameworks against criminal ground truth outside a high-income setting. Classical screens and the Kawai RD yield near-null AUCs; worker-flow screens yield a null after temporal conditioning. The benchmark itself—linking CADE rulings, BEC bid-level records, and RAIS employer–employee data—is a reusable infrastructure for future detection research.

The remainder of the paper proceeds as follows. Section 2 situates our work in the detection, network, and worker-mobility literatures. Section 3 describes public procurement in São Paulo and the CADE cartel cases that anchor the ground truth. Section 4 presents the empirical strategy—including the temporal-conditioning test that distin-

guishes prospective from retrospective signal—and a framework for screen heterogeneity across market structures. Section 5 describes the data sources and sample construction. Section 6 reports the results in two parts: the apparent detection result and its temporal unraveling. Section 7 discusses implications and concludes.

2. Related Literature

Our paper sits at the intersection of three literatures: the design and validation of cartel-detection screens, the use of network structure to identify collusion, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

The idea that bid-rotation agreements leave statistical footprints is now well established. Porter and Zona (1993) showed that collusive phases of the 1880s railroad freight cartel exhibit systematically lower bid dispersion than competitive phases, exploiting the cartel's own internal records as an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement; Bajari and Ye (2003) proposed exchangeability tests for bidder costs in Midwestern highway paving; Conley and Decarolis (2016) extended pair-level correlation tests to Italian procurement. Imhof et al. (2018) demonstrated that combining multiple screens in a portfolio outperforms any single indicator in Swiss data, and Abrantes-Metz et al. (2006) offered practical guidance on implementing variance screens while cautioning against single-indicator reliance. Harrington (2008) synthesized the design principles, emphasizing a point our findings will make concrete: the informativeness of any screen depends on the competitive counterfactual *and* the coordination technology the cartel actually uses.

A complementary strand focuses on the bid margin itself. Kawai et al. (2023) proposed a regression discontinuity centered on the winning–losing threshold in Japanese construction procurement: under genuine competition, narrowly winning and narrowly losing firms should look alike, and any sharp discontinuity identifies a non-competitive wedge. Chassang and Ortner (2019) developed a theoretical framework in which bid constraints endogenously shape the collusive technology—predicting, among other things, that format-specific screens may outperform generic variance tests, a prediction consistent with the format heterogeneity we document below.

A common thread across these contributions is that nearly all validation exercises rely on ground truth from high-income economies with deep bidder pools and well-resourced enforcement agencies. Levenstein and Suslow (2006) documented across 81 international cartels that market structure, transparency, and enforcement capacity jointly determine cartel success—suggesting that screens validated in one institutional regime need not travel well to another. Direct validation against criminal ground truth in a middle-income setting with structurally different labor markets remains scarce, and that is the gap our conviction-anchored benchmark is designed to fill.

2.2. Network-based and machine-learning approaches

A parallel literature has moved beyond univariate bid statistics to exploit the relational structure of procurement markets. Wachs and Kertész (2019) showed that co-bidding networks in Georgian procurement flag candidate cartels through group-level cohesion and exclusivity, even without criminal ground truth. Signor et al. (2020) built a probabilistic detector from the data released by Brazil's *Operação Lava Jato*, reporting 81–96 percent accuracy on the Petrobras construction cartel. Huber and Imhof (2019) combined classical screens with supervised machine-learning classifiers in Swiss data, correctly predicting 84 percent of collusive tenders out of sample. Clark et al. (2018) documented how collusive networks in Québec construction co-evolve with political connections, illustrating the interplay between bid rigging and corruption.

A more recent wave extends the network logic to data sources *outside* the bid record. Asai and Charoenwong (2026) link shareholder registries to procurement auctions in Singapore and show that common-ownership links predict identical bids; removing the ownership-network effect would cut government spending by 4–6 percent. Villamil et al. (2024) build temporal multiplex networks that combine ownership ties and co-bidding patterns in Swedish procurement, finding that denser ownership connections predict higher single-bidding and more missing bidders. Together, these papers establish that external network layers—corporate ownership, in their case—carry detection signal that co-bidding statistics alone cannot reach.

We add a third layer: labor-market connectivity, constructed from matched employer–employee records. Worker-flow networks differ from ownership networks in two ways that matter for detection. They capture *operational* coordination—personnel rotation, knowledge transfer, shared technical staff—rather than financial ties, and thus speak to a distinct coordination channel. And they are available wherever governments maintain

compulsory employer–employee registries (RAIS in Brazil, DADS in France, IAB in Germany), making the approach portable without new data collection. Because worker-flow data are informationally independent of both the bid record and the corporate registry, the risk of circularity between the screen and the behavior it targets is lower than for co-bidding or ownership-based approaches.

2.3. Worker mobility as an inter-firm channel

Labor economists have long recognized that workers carry knowledge between firms. Stoyanov and Zubanov (2012) showed that productivity gains spill across Danish firms through the mobility of their employees, with larger spillovers when the productivity gap between sender and receiver is wide. Serafinelli (2019) documented a similar pattern in Italy's Veneto region: workers arriving from high-productivity firms raise wages and output at their new employers. Jarosch et al. (2021) developed an equilibrium model in which team composition and worker reallocation jointly shape firm-level human capital. The upshot is that worker flows are not just a labor-market phenomenon—they are an active conduit for inter-firm information transmission.

Three recent papers connect this conduit to collusion. Lamichhane and McGee (2024) provide experimental evidence that, in Bertrand markets, worker mobility between firms facilitates collusive pricing: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices and more market sharing, especially in concentrated markets. Herrera-Caicedo et al. (2025) study product-market collusion facilitated by shared directors and executives. They find that common leadership raises the probability of collusion by 11 percentage points—and, strikingly, that inter-firm worker flows *fall* by roughly 20% once a board interlock is established. The substitution is telling: when collusion runs through leadership ties, personnel-flow coordination becomes redundant. Our institutional-thinness model predicts the mirror image: when leadership ties are absent and coordination relies on operational personnel exchange, worker-flow screens should carry the signal that ownership and board-interlock screens miss.

On the data-infrastructure side, two papers have already linked RAIS to Brazilian procurement—though not for detection. Barbosa and Straub (2020) track worker movements between public agencies and private suppliers, showing that supplier-to-administration revolving doors are detrimental to procurement efficiency. Szerman (2023) exploits a debarment-policy change to estimate the worker-level costs of anti-corruption enforcement, finding that blacklisted firms lose nearly half their employees.

Both papers prove the RAIS–procurement linkage is feasible and productive; neither uses it as a cartel screen.

We bring these threads together. Lamichhane and McGee (2024) establish the mechanism—worker mobility as a collusion channel. Asai and Charoenwong (2026) and Villamil et al. (2024) show that external network layers carry detection signal. Barbosa and Straub (2020) and Szerman (2023) demonstrate the data infrastructure. What has been missing is a test of whether the mechanism translates into a working screen—and, more fundamentally, whether the signal that network screens capture in panel data reflects prospective coordination or retrospective enforcement effects. None of the studies above conditions the network measure on temporal precedence: the full employment, ownership, or co-bidding panel is used to construct the screen, and the AUC is evaluated against convictions that occurred during or before the panel period. If enforcement changes the network—through labor reallocation, ownership restructuring, or altered bidding patterns—the screen may absorb these aftermaths and overstate its real-time usefulness. This paper provides the first test of that distinction.

3. Institutional Background

3.1. Public procurement in São Paulo: the BEC platform

São Paulo's state government purchases common goods and services through a decentralized network of roughly 1,300 public buyer units (PBUs), all of which have been required since 2007 to transact on the state's electronic platform, the *Bolsa Eletrônica de Compras* (BEC). The scale is substantial: in 2018 alone, BEC handled approximately US$1.7 billion in procurement, and over the study period the platform processed more than R$7.4 billion for the State Health Secretariat and other agencies.

The BEC catalog organizes items by group and class—from pharmaceuticals to office supplies—totaling about 6,350 distinct product codes. Each tender, called a purchase offer (PO), may contain multiple items; each item is auctioned separately, so a unique procurement event is defined by a (PO $\times$ item code) pair, which we label POI. The platform records everything a researcher needs: the reference price set by the buyer, every bid submitted, the identities of bidders and the winning firm, and the auction format.

Two formats predominate. In the *convite* (sealed-bid tender), firms submit confidential bids by a deadline; the lowest qualified bid wins. This format is typically used

for purchases below R$176,000 (about US$35,000). In the *pregão eletrônico* (descending auction), firms first submit sealed proposals, after which an open phase begins: bidders observe the current lowest price in real time and can underbid for an initial 20 minutes, with automatic 4-minute extensions whenever a late bid arrives. The auction closes after 4 consecutive minutes of silence. The pregão carries no value ceiling and accounts for the majority of BEC spending.

The distinction matters for collusion. The pregão's iterative, anonymous bidding gives cartel members the ability to monitor compliance in real time but also exposes deviations; the convite's single-shot format makes monitoring harder but bid coordination simpler. Our primary analysis focuses on pregão auctions, for which the close-margin RD framework is most naturally motivated; convite results are reported as a complementary check.

3.2. Antitrust enforcement: CADE and the ground truth

Brazil's federal competition authority, the *Conselho Administrativo de Defesa Econômica* (CADE), is an autonomous agency within the Ministry of Justice. Its Tribunal renders administrative decisions—binding fines and behavioral remedies, not criminal sentences in the judicial sense. Cases typically begin with a complaint or a leniency application (*Termo de Compromisso de Cessação*, TCC), proceed through a formal investigation by the *Superintendência-Geral*, and conclude with a Tribunal vote.

Our ground truth draws on CADE proceedings in which the Tribunal convicted firms of cartel conduct in São Paulo procurement markets. We searched CADE's public repository for proceedings with final decisions that (i) name specific firms with identifiable CNPJs, (ii) specify the product market, and (iii) delineate a conduct period. Seven cartels meet these criteria, spanning sectors as varied as pharmaceuticals, school meals, trash bags, solar water heaters, and metropolitan rail, and involving 35 firms whose CNPJ roots we can match to BEC records.

What makes these case files particularly useful is that they specify not just *who* colluded but *when* and *in which market*. We exploit this granularity to label individual auctions as cartel-tainted—matching each convicted firm's CNPJ root, the CADE-specified conduct window, and the relevant product code to BEC tenders. The result is a benchmark that captures both the identity and timing dimensions of collusion, rather than treating every auction involving a flagged firm as equally suspect.

4. Empirical Strategy

We bring five detection approaches to the conviction-anchored benchmark: a close-margin regression discontinuity (Section 4.1), classical bid-distribution screens (Section 4.2), network screens built from worker-flow and co-bidding data (Section 4.3), a specialization placebo (Section 4.6), and—crucially—a temporal-conditioning test that distinguishes prospective from retrospective detection power (Section 4.5). Every screen is evaluated by its area under the ROC curve (Section 4.4). A simple model in Section 4.7 organizes predictions about where classical screens should fail and where post-enforcement labor reallocation should concentrate.

4.1. Close-margin regression discontinuity

Following Kawai et al. (2023), we focus on auctions decided by a hair. Define the margin of victory in auction a as

$$MV_a = \frac{b_a^{\text{runner-up}} - b_a^{\text{win}}}{b_a^{\text{win}}}.$$

When MV_a is close to zero, the winner barely underbid the nearest rival. Under genuine competition, narrowly winning and narrowly losing firms should look alike on predetermined characteristics; a sharp discontinuity at the cutoff signals something non-competitive.

We estimate the jump using local-polynomial regression with a data-driven bandwidth. Two outcomes carry interpretive weight. *Incumbency*—whether the firm won the previous auction in the same market—tests for bid rotation: a jump would indicate that designated winners are systematically past winners. *Last bidder*—whether the firm submitted the final bid—tests for strategic timing. A density test on the running variable checks whether the margin itself is being manipulated.

4.2. Classical bid-distribution screens

We compute four screens on firm-best bids (one bid per firm per auction, keeping the lowest when multiple rounds exist):

1. **Coefficient of variation** (CV_a): standard deviation divided by mean. Bid rotation predicts lower CV because cover bids cluster near the designated winner (Porter and Zona, 1993; Abrantes-Metz et al., 2006).

2. **Bid spread** (S_a): (max − min)/mean. Like CV, compressed under cover bidding.
3. **Margin of victory** (MV_a): the designated winner should undercut cover bids by just enough to win, yielding a small margin.
4. **Number of bidders** (N_a): entry deterrence or market division should reduce the count (Marshall and Marx, 2012).

For each, we compare cartel-tainted auctions to clean controls on the close-margin benchmark ($|MV| \leq 0.10$), reporting the Welch t-statistic, Cohen's d, the Wilcoxon rank-sum p-value, and the AUC.

4.3. Network-based screens

We step outside the bid data and build three screens from administrative sources.
Worker-flow screens. For each (winner, runner-up) pair we measure labor connectivity from the full 2009–2017 RAIS panel:

- **Shared workers** (W_{ij}): the count of distinct PIS numbers appearing at both firms during the nine-year window.
- **Jaccard similarity** ($J_{ij} = W_{ij}/(N_i + N_j - W_{ij})$): normalizes the count by firm size, so that large employers are not mechanically flagged.

The logic is straightforward. If cartel coordination in thin labor markets runs through personnel—managers carrying pricing information, technicians rotating between member firms—then the employment records should show it (Stoyanov and Zubanov, 2012; Serafinelli, 2019). Alternatively, if the signal is retrospective—reflecting labor reallocation *after* the cartel period ends—it would still appear in the full-panel measure but would vanish under temporal conditioning. We test both possibilities.
Co-bidding screen. For each pair we count the auctions in which both firms submitted bids (C_{ij}). This provides a network channel built entirely within the procurement data.

4.4. Detection evaluation: AUC

We use the area under the ROC curve to compare screens. The AUC is simply the probability that a randomly chosen cartel auction scores more "cartel-like" than a randomly chosen clean auction. For screens where lower is more suspicious (CV, S, MV, N), we negate scores so that AUC > 0.5 always means detection in the predicted

direction. We compute the AUC via the Mann–Whitney U statistic with average-rank tie-handling:

$$\text{AUC} = \frac{U_1}{n_1 \times n_0},$$

where n_1 and n_0 are the cartel and clean sample sizes. All screens are evaluated on the same unified close-margin benchmark, so comparisons are apples-to-apples. Appendix A.1 shows that results are stable across bandwidths from 0.02 to 0.50.

4.5. Temporal conditioning and decomposition

Network screens built from administrative panel data face a fundamental identification challenge: the measure uses information from the *entire* panel, including periods after the auction being screened. If cartel detection or enforcement changes the network—for instance, by triggering labor reallocation from convicted firms to competitors—then the full-panel AUC conflates prospective detection power with retrospective correlation. We implement two tests.

Temporal conditioning. For each auction at year t, we recompute the worker-flow and co-bidding screens using only data from years strictly before t. If the AUC survives, the screen captures information available at the time of the auction. If it collapses, the signal is retrospective. We report a cumulative backward window $[2009, t-1]$ and a rolling window $[\max(2009, t-3), t-1]$.

Temporal decomposition. For each shared worker between a cartel firm and a rival, we classify the inter-firm link by when it was first observable relative to the cartel's conduct period: **pre-conduct** (both firms employed the worker before the start year), **during-conduct** (the link appeared during $[\text{start}, \text{end}]$), or **post-conduct** (the link appeared after the end year). We then use each window's shared-worker count as a separate screen and evaluate it against the benchmark. This decomposition identifies *when* the signal arises—and whether it reflects coordination, ongoing cartel activity, or enforcement aftermath.

4.6. Specialization placebo

Elevated worker flows between a cartel firm and its rival could reflect several forces: pre-existing personnel connections, the natural circulation of a thin labor pool, or post-conduct labor reallocation. To separate market-level circulation from the cartel-specific component, we build a three-way comparison. The **cartel** sample consists of auction-level (winner, runner-up) pairs with at least one cartel-active firm ($N \approx 2{,}300$). The

placebo consists of non-cartel pairs bidding on the same BEC item code in the same year. The **clean** baseline is every other non-cartel pair.

If the entire worker-flow signal were market specialization, the placebo should match the cartel sample. Any excess in the cartel sample over the placebo is a cartel-specific residual—though, as we show in the results, this residual is driven primarily by post-conduct displacement rather than by coordination during the cartel period.

4.7. A framework for screen heterogeneity

Why should screen effectiveness depend on the market? And why should post-enforcement labor reallocation concentrate in thin markets? Consider $n \geq 2$ firms operating in a product market with specialized-labor pool of thickness L.

The first question has a standard answer. Classical bid-distribution screens exploit the compression that bid rotation imposes on the bid distribution. In thick markets (large L, many bidders), rotation is the natural coordination device and the compression is detectable. In thin markets (small L, few bidders), the bid distribution is inherently noisy—few firms generate high variance even under collusion—and phantom-bidding or cover-bid mechanisms may inflate rather than compress dispersion. Classical screens should therefore work in thick markets and fail in thin ones.

The second question is about the labor-market aftermath of enforcement. When a cartel firm contracts—losing contracts, facing fines, undergoing investigation—it sheds workers. In thick labor markets, those workers disperse across many potential employers; the per-firm flow is dilute and hard to detect. In thin markets, the same workers have few alternative employers, and the flows concentrate in the small set of firms that operate in the same specialized product market. The worker-flow screen, evaluated on the full panel, captures this concentration—but it is a concentration of *displacement*, not of *coordination*.

Three testable predictions follow:

1. **Thickness gradient for classical screens.** In thin markets, classical AUCs should be near or below chance. In thick markets, they should recover. This prediction is independent of the temporal-conditioning critique and uses contemporaneous bid data.
2. **Thickness gradient for post-conduct reallocation.** Full-panel worker-flow AUCs should be highest in thin markets—not because coordination runs through

personnel, but because displacement concentrates where the labor pool is shallowest.

3. **Temporal asymmetry.** If the worker-flow signal reflects coordination, it should be present in pre-conduct flows. If it reflects enforcement aftermath, it should be concentrated in post-conduct flows. The temporal decomposition directly tests this.

The framework is deliberately spare. It abstracts from repeated-game dynamics, leniency, and heterogeneity across auction formats. We offer it not as a structure to be estimated but as a lens for organizing the results and for specifying *when* each family of screens should and should not carry signal.

5. Data

The analysis rests on three administrative data sources—procurement bids, employment records, and antitrust rulings—which we link at the firm level to construct the conviction-anchored benchmark.

5.1. BEC: bid-level procurement data

We draw on the complete bid-level records from São Paulo's *Bolsa Eletrônica de Compras* (BEC) for January 2009 through December 2019. After restricting to pregão and convite procedures with at least two participating firms, the data contain 2,613,010 competitive auctions, 13,957,419 firm-level bid participations, 40,273 distinct firms, and 153,048 unique item codes. Each record identifies the purchase offer, item code, participating firm (by CNPJ), bid amount, outcome, auction format, and purchasing agency.

For each auction we compute the screens described in Section 4.2—coefficient of variation, bid spread, margin of victory, and bidder count—on firm-best bids (keeping each firm's lowest bid when multiple rounds exist). Markets are defined transitively following Kawai et al. (2023): two firms share a market if they ever bid on the same item code in the same auction. This yields 7,581 distinct markets.

5.2. RAIS: employer–employee matched panel

The *Relação Anual de Informações Sociais* (RAIS) is Brazil's compulsory annual census of formal employment. Every establishment with a CNPJ must report each employment bond active at any point during the year, identified by the worker's PIS/PASEP

number. We use the 2009–2017 vintages, restricting to establishments whose 8-digit CNPJ root appears in the BEC firm universe—roughly 19,000 roots. The result is a nine-year panel of every formal-sector job at every BEC-active firm.

From this panel we build the worker-flow edge list. For each pair of BEC firms (i, j), we count distinct PIS numbers that appear at both firms at any point during the nine years (the shared-worker count W_{ij}) and compute the Jaccard similarity J_{ij}. Pairs with no shared workers are zero-imputed when merged to the auction benchmark.

5.3. CADE: cartel ground truth

Our ground truth comes from CADE administrative proceedings with final Tribunal decisions convicting named firms of cartel conduct in São Paulo procurement markets (see Section 3). Table 1 lists the seven cartels we identified—spanning pharmaceuticals, school meals, trash bags, solar water heaters, metropolitan rail, airport cafeterias, and school transport—together with their conduct periods and the 35 firms whose CNPJ roots match BEC records.

The key feature for detection is that each case file specifies not just the colluding firms but the product market and time window. We exploit this to label individual auctions as cartel-tainted: an auction earns the flag only if a convicted firm bid during the CADE-specified conduct period for the relevant product and year. This firm $\times$ period $\times$ product matching avoids the coarser alternative of treating every auction involving a flagged firm as suspicious.

5.4. Sample construction

All detection tests use the close-margin envelope $|MV| \leq 0.10$. Within that envelope we define three samples:

1. **Cartel-active benchmark** ($N \approx$ 2,300 pregão; $N = 263$ convite). At least one convicted firm bid during the active period of its cartel. This is the primary evaluation sample.
2. **Strict pair-matched sample** ($N = 10$ pregão). Both the winner and the runner-up belong to the same convicted cartel during the same conduct period. The gold standard for head-to-head coordination, but power-limited.
3. **Anytime-listed sensitivity** ($N \approx 80$ pregão). Both firms appear in a CADE conviction for the same cartel, but the auction year may fall outside the conduct

Table 1: Convicted Cartels in São Paulo Procurement Markets

Sector	CADE process	Period	Firms	In BEC	Conduct type
Trens e metrôs	08700.004617	1999–2013	13	10	Bid rigging
Merenda escolar	08012.010022	2006–2013	7	6	Market division
Medicamentos	08012.002222	2007–2011	8	8	Hub-and-spoke
Sacos de lixo	08700.005789	2008–2014	3	3	Bid rigging / price fixing
Aquecedores solares	08012.001273	2009–2010	5	4	Bid rigging
Cafeterias Infraero	08700.007278	2014	6	3	Bid rigging
Transporte escolar	08700.005876	2019	2	2	Bid rigging
Total (unique CNPJ roots)			35	36	

Notes: "Firms" and "In BEC" count unique 8-digit CNPJ roots per cartel; row totals are after cross-cartel deduplication. The In BEC total exceeds the Firms total because one firm participates in two cartels. Conduct periods hand-curated from CADE Tribunal decisions and press releases (see Appendix). Source: script 12.

window. Of the 107 pre-filtering auctions, 93 are out-of-window and 100 come from a single case (the medicamentos hub-and-spoke, CADE 08012.002222/2011-09). Reported as a sensitivity only.

The *clean control* is every close-margin auction with no cartel-listed firm (N = 527,259 pregão; 519,557 convite). All screens are evaluated against this common baseline.

5.5. Variable definitions

Table 2 collects the key variables.

6. Results

We present the results in two acts. The first (Sections 6.1–6.3) establishes the facts: classical screens fail, and worker-flow screens appear to succeed. The second (Sections 6.4–6.7) tests whether the apparent success is genuine—and discovers that it is not. The remaining subsections interpret the pattern that emerges.

6.1. Classical screens and the close-margin RD

Table 14 presents the unified benchmark comparison of all screens on the same close-margin pregão sample ($|MV| \leq 0.10$). Panel A reports the classical bid-distribution

Table 2: Variable Definitions

Variable	Definition
MV_a	Margin of victory: $(b_a^{\text{runner-up}} - b_a^{\text{win}})/b_a^{\text{win}}$
CV_a	Coefficient of variation of firm-best bids
S_a	Bid spread: $(\max - \min)$/mean of firm-best bids
N_a	Number of distinct firms submitting bids
Incumbent_a	$= 1$ if the winning firm won the preceding auction in the same market
Last bidder_a	$= 1$ if the winning firm submitted the final bid
W_{ij}	Shared workers: distinct PIS appearing at both firms i and j
J_{ij}	Jaccard similarity: $W_{ij}/(N_i + N_j - W_{ij})$
C_{ij}	Co-bidding count: auctions where both i and j bid

screens on the cartel-active benchmark (N = 2,319 cartel-tainted auctions vs. N = 527,259 clean controls).

All four classical screens fire in the direction predicted by the bid-rotation model: the coefficient of variation ($\Delta = -0.035$, $d = -0.07$), bid spread ($\Delta = -0.173$, $d = -0.10$), and number of active bidders ($\Delta = -0.87$, $d = -0.24$) are compressed at cartel-tainted auctions relative to clean controls. Welch t-statistics are large and statistically significant, owing to the massive control sample. Yet the effect sizes are small: Cohen's d ranges from -0.07 to -0.24, indicating near-total distributional overlap between cartel and clean auctions. The AUCs reflect this overlap, ranging from 0.52 (coefficient of variation) to 0.56 (number of bidders)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation ($p < 10^{-4}$ for each screen, consistent with the AUC direction).

The margin of victory exhibits an anomalous pattern: $\Delta MV = +0.004$ ($d = +0.17$), with an AUC of 0.46—slightly *below* chance and in the *opposite* direction to the bid-rotation prediction. Cartel-active auctions have marginally *larger* winning margins, not smaller. We interpret this as consistent with a partial phantom-bidding mechanism in which cover bids are inflated relative to the designated winner's price, widening rather than compressing the margin.

A direct replication of the Kawai et al. (2023) close-margin RD detects a baseline incumbency advantage on the full sample ($\hat{\beta} = 0.018$, SE $= 0.005$, $p < 0.001$), reflecting a competitive pattern in which recent winners are slightly more likely to win again. On

the cartel-active subsample, however, no *additional* discontinuity is detected ($p > 0.15$): cartel exposure does not amplify the baseline incumbency pattern, nor does it produce a detectable jump in last-bid behavior. A density test detects no manipulation of the running variable ($T_{\text{CJM}} = -0.02$, $p = 0.98$). Neither the RD design nor the density test flags the cartel-active auctions as anomalous relative to competitive bidding.

6.2. Heterogeneity by coordination technology

The aggregate null conceals a sharp split by coordination technology. Table 3 disaggregates the benchmark by the conduct type documented in CADE's proceedings: five cartels operated through *bid rigging* (with or without price fixing or market division), while one—the medicamentos hub-and-spoke cartel—coordinated through a central distributor.

Table 3: Classical Screen AUCs by Coordination Technology

	N	CV	Spread	N_{firms}
Bid rigging (5 cartels)	342	**0.652**	**0.654**	**0.600**
Hub-and-spoke (1 cartel)	1,977	0.501	0.510	0.547
All (benchmark)	2,319	0.523	0.531	0.555

Notes: Pregão, $|MV| \leq 0.10$. Bid rigging includes *trens e metrôs*, *merenda escolar*, *sacos de lixo*, *transporte escolar*, and *cafeteria aeroporto*. Hub-and-spoke is the *medicamentos* cartel (CADE 08012.002222/2011-09). Classical screen AUCs use "lower is cartel" convention. Source: script 34.

Classical screens detect bid rigging. On the 342 cartel-active auctions involving bid-rigging cartels, the CV AUC reaches 0.65, spread reaches 0.65, and bidder count reaches 0.60—all substantively above chance and operationally useful as triage indicators. On the 1,977 hub-and-spoke auctions, every classical screen is at or near chance (0.50–0.55). The aggregate null of 0.52–0.56 is a composition effect: the hub-and-spoke cartel, which accounts for 85% of the sample and leaves no trace in bid patterns, dilutes the bid-rigging signal that would otherwise be visible.

The result makes economic sense. Bid-rigging cartels coordinate by arranging cover bids around a designated winner, which compresses the bid distribution—exactly the signature that variance screens are designed to detect. Hub-and-spoke cartels coordinate through a central intermediary (in this case, a pharmaceutical distributor) that sets prices upstream; individual firms may submit bids that look competitive at the auction level even when prices are pre-arranged at the wholesale level. Classical screens, which

operate on the bid distribution, are architecturally incapable of detecting coordination that occurs outside the auction.

This finding is the paper's positive detection result. It does not require temporal conditioning—it uses contemporaneous bid data—and it does not depend on the medicamentos cartel that dominates the benchmark. The message for competition authorities is direct: classical variance screens detect bid rigging at operationally useful levels (AUC ≈ 0.65), but they are blind to hub-and-spoke coordination. Matching the screen to the suspected coordination technology is essential.

6.3. Network-based screens: an apparent breakthrough

If classical screens detect bid rigging but miss hub-and-spoke, can network screens fill the gap? Panel B of Table 14 turns to the network screens, evaluated on the same $N = 2{,}319$ cartel-active auctions.

The answer appears to be worker flows. The shared-worker count yields an AUC of 0.750 ($\Delta = +3.70$, $t = +14.1$), and the Jaccard similarity yields an AUC of 0.739 ($\Delta = +0.0014$, $d = +0.19$, $t = +20.7$). Both outperform the best classical screen (number of bidders, AUC = 0.56) by roughly 20 percentage points. A third screen constructed from co-bidding density yields an AUC of 0.695, providing independent corroboration through a distinct information channel.

The signal is not an artifact of sectoral composition. Restricting attention to firm pairs that share the same 2-digit CNAE sectoral classification reduces the Jaccard AUC only marginally, from 0.739 to approximately 0.730. The residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline.

Decomposing the AUC by cartel role reveals symmetry. Of the 2,319 cartel-active auctions, 996 have the winner as the cartel-active firm, 1,313 have the runner-up as the cartel firm, and only 10 have both. The AUC is stable across these subgroups (0.74–0.76), indicating that the signal does not depend on whether the cartel firm wins or loses.

These results appear compelling. But they rest on a critical assumption: that the worker-flow measure captures information available at the time of the auction. The shared-worker count uses the full 2009–2017 RAIS panel—which means that for an auction held in 2010, the screen incorporates worker movements that occurred in 2015. Two features of the benchmark compound this concern. First, one cartel (medicamentos,

conduct 2007–2011) accounts for 85% of the cartel-active sample, so the headline AUC is overwhelmingly a statement about that single case. Second, in 99.6% of cartel-active auctions only one firm in the (winner, runner-up) pair is a convicted cartelist—the screen is measuring the connectivity of a cartel firm with an arbitrary non-cartel rival, not the coordination between two co-conspirators. We turn next to the question that these observations make unavoidable: does the signal survive when we restrict the employment data to information that was actually available before each auction?

6.4. Temporal conditioning: the signal collapses

For each auction at year t, we count shared workers using only RAIS observations from years strictly before t: the cumulative backward window $[2009, t-1]$. This preserves the full employment history available at the time of the auction while eliminating any look-ahead. The resulting AUC is 0.52—indistinguishable from chance. A three-year rolling window $[t-3, t-1]$ produces an essentially identical result (0.54). Table 4 summarizes.

Table 4: Worker-Flow AUC under Temporal Conditioning

Window	AUC	N_{cartel}
Full panel $[2009, 2017]$	**0.750**	2,319
Cumulative backward $[2009, t-1]$	0.524	2,319
Rolling $[t-3, t-1]$	0.536	1,826

Notes: Pregão, $|MV| \leq 0.10$. Full panel uses all RAIS 2009–2017 observations regardless of auction year. Cumulative backward restricts to RAIS years $< t$ for auction year t. Rolling uses $[\max(2009, t-3), t-1]$ and restricts to auctions in 2010–2017. AUC computed via Mann–Whitney U (shared workers, higher is cartel). Sources: scripts 23, 25, 26.

The gap between 0.75 and 0.52 is the central fact of this paper. The prospective detection power of the worker-flow screen is zero on this benchmark. The signal that produces the headline result comes entirely from worker movements that occurred *after* the auction being screened.

The bias is not specific to worker-flow screens. The co-bidding density measure, which achieves an AUC of 0.70 on the full panel (Table 14), falls to 0.55 under cumulative temporal conditioning—a drop of 0.15, compared with 0.23 for shared workers. Both network screens lose their detection power once future information is excluded.

6.5. Temporal decomposition: the signal is post-conduct

If the worker-flow signal is retrospective, where exactly does it come from? We classify each shared worker by when the inter-firm link was first observable relative to the cartel's conduct period:

- **Pre-conduct**: both firms employed the worker before the cartel's start year.
- **During-conduct**: the link became observable during the conduct period [start, end].
- **Post-conduct**: the link became observable after the conduct period ended.

Table 5: AUC by Temporal Window of Worker Flow

Window	AUC	Mean (cartel)	Mean (clean)
Pre-conduct	0.500	0.000	0.000
During-conduct	0.618	2.20	0.00
Post-conduct	**0.773**	3.47	0.00
All windows	0.789	5.67	0.00

Notes: Each row uses the shared-worker count from the indicated temporal window as the screen. Pregão, $|MV| \leq 0.10$, $N_{\text{cartel}} = 2{,}319$, $N_{\text{clean}} = 527{,}259$. Source: script 27.

The decomposition in Table 5 is definitive. Pre-conduct worker flows produce an AUC of exactly 0.50: there is no detectable difference in pre-existing personnel connections between cartel-active and clean auction pairs. During-conduct flows produce a modest 0.62. Post-conduct flows produce 0.77—nearly the full headline result. The signal is an aftermath, not a precursor.

The post-conduct flows are directional. Across all cartels, 30,977 distinct workers moved from cartel firms to rivals after the conduct period, versus 13,236 in the reverse direction—a 2.3:1 ratio. The asymmetry intensifies over time: in 2010, the cartel-to-rival ratio is 1.4:1; by 2016–2017, it reaches 4:1. The pattern is consistent with enforcement-driven labor displacement: as CADE investigations proceed and conduct periods end, convicted firms shed employees who are absorbed by competitors in the same thin labor markets.

6.6. Event study: do outflows spike after conduct ends?

The temporal decomposition shows *where* the signal lives (post-conduct flows), but not whether outflows *increase* relative to a counterfactual. To test this, we estimate a

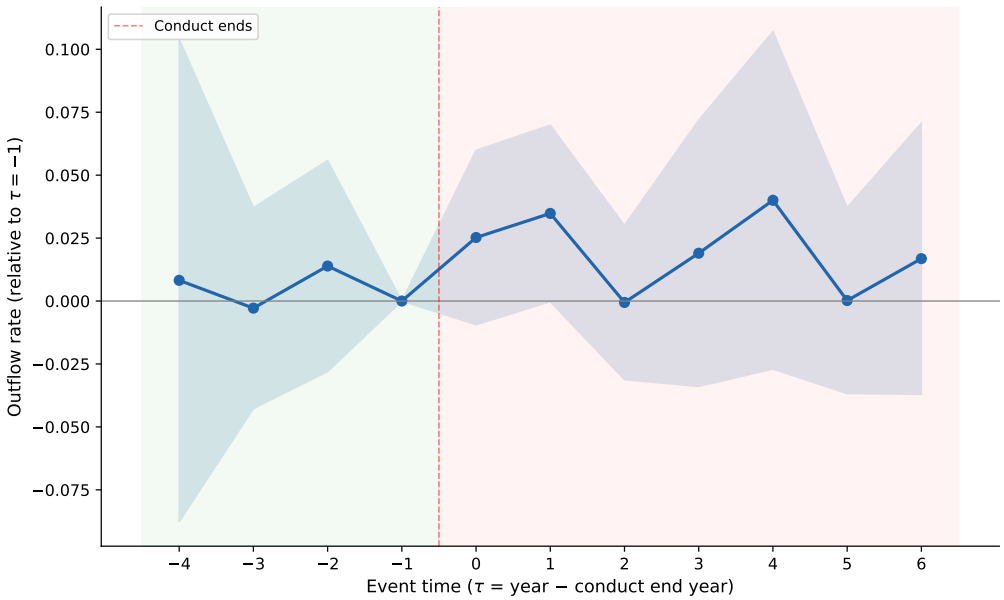
firm-level event study around the end of each cartel's conduct period:

$$\text{outflow_rate}_{it} \;=\; \alpha_i + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \cdot D_{i\tau} + \varepsilon_{it}\,,$$

where the outcome is the share of firm i's workers in year $t-1$ who appear at a different BEC firm in year t, α_i and γ_t are firm and year fixed effects, and $D_{i\tau}$ indicates that firm i is a cartel firm at event time $\tau = t - \text{end_year}_i$. The omitted category is $\tau = -1$. Standard errors are clustered at the firm level. The control group is all non-cartel BEC firms with at least three employees, yielding a panel of 159,140 firm-year observations (26,928 firms, 257 treated observations from 41 cartel firms with conduct ending by 2017).

Figure 1 plots the event-study coefficients. Two patterns stand out. *First*, pre-treatment coefficients are flat and insignificant (max $|t| = 0.65$ for $\tau \in [-4, -2]$): cartel and control firms follow parallel outflow trajectories before the conduct period ends. *Second*, outflows rise after the conduct period ends. The post-treatment coefficients average +0.019, a 29% increase relative to the control-group mean outflow rate of 6.7%. The effect peaks at $\tau = +1$ ($\hat{\beta} = +0.035$, $t = 1.94$) and $\tau = +4$ ($\hat{\beta} = +0.040$, $t = 1.17$).

Figure 1: Event Study: Worker Outflows around Cartel Conduct End



Notes: Each point is the coefficient $\hat{\beta}_\tau$ from the TWFE specification above, with 95% confidence intervals (firm-clustered SEs). $\tau = 0$ is the first year after conduct ends. The dashed vertical line marks the conduct-end date. Shading distinguishes pre-treatment (left) from post-treatment (right). The control group is all non-cartel BEC firms with ≥ 3 employees. Source: script 28.

The estimates are imprecise—only $\tau = +1$ is marginally significant at the 10% level—reflecting the small number of treated firms (41). But the direction is unambiguous, the pre-trends are clean, and the magnitude (+29%) is economically meaningful. The event study provides suggestive evidence that the post-conduct worker-flow pattern documented in Section 6.5 reflects a genuine increase in outflows from cartel firms, not a pre-existing difference in mobility rates.

6.7. Where do the workers go?

Who absorbs the displaced workers—the cartel firm's direct auction competitors, or unrelated firms? We classify each destination into three categories: **auction rivals** (318 firm pairs that were runner-up or winner against the cartel firm in a close-margin auction during conduct), **market peers** (2,181 pairs that bid on the same item codes but were never in a close-margin pair), and **bystanders** (all other BEC firms).

Table 6: Post-Conduct Worker Flows by Rival Type

Destination type	Workers	Share	Dest. firms	Workers/firm
Auction rivals	382	1.0%	44	8.7
Market peers	5,606	14.5%	183	**30.6**
Bystanders	32,663	84.5%	3,903	8.4
Total	38,651	100%		

Notes: Post-conduct flows from cartel firms (conduct end $\leq$ 2017) to non-cartel BEC firms. Auction rivals defined from close-margin pregão pairs ($|MV| \leq 0.10$) during the conduct period. Source: script 31.

The bulk of the reallocation (84.5%) goes to bystanders, consistent with generic labor-market mobility. But the *intensity* tells a different story: market peers receive 30.6 workers per destination firm, 3.6 times the bystander rate. Workers from cartel firms disproportionately concentrate in firms operating in the same product markets, consistent with sector-specific human capital flowing through the only available channel in thin labor markets.

A direct test of pair-level absorption reinforces this pattern. Of 993 close-margin auctions where the cartel firm won during conduct, 463 (46.6%) show at least one ex-worker of the winning cartel firm subsequently appearing at the specific runner-up of that auction. Nearly half of all close-margin winner–runner-up pairs exhibit a post-conduct worker transfer—not mass displacement along auction lines, but a fine-grained pattern

of specialized workers finding their way to the small set of firms in the same thin product market.

6.8. Who moves? Worker characteristics

If the post-conduct flows carry cartel-specific knowledge—pricing strategies, coordination protocols—we would expect managers, professionals, and high-wage workers to be overrepresented among movers. Table 7 compares three groups: workers who leave cartel firms for other BEC firms post-conduct (*cartel movers*, $N = 17{,}381$), workers who remain at cartel firms (*cartel stayers*, $N = 47{,}255$), and workers who move between non-cartel BEC firms in the same period (*control movers*, $N = 296{,}036$, 10% sample).

Table 7: Characteristics of Workers Who Leave Cartel Firms

	Cartel movers	Cartel stayers	Control movers
Continuous variables (means)			
Wage (minimum wages)	2.4	3.9	2.2
Tenure (months)	31.7	38.9	26.6
Age	37.2	37.2	33.6
Occupation (%)			
Directors & managers	2.6	3.5	3.3
Professionals (degree-required)	7.3	9.3	9.1
Service & sales	**55.1**	36.8	37.9
Production & craft	10.5	21.5	12.9
Demographics (%)			
Male	40.0	58.8	62.5
Tertiary education	14.6	17.9	17.6

Notes: Characteristics measured in the last year of the cartel's conduct period. Cartel movers: workers at a cartel firm who subsequently appear at a different BEC firm. Cartel stayers: workers who remain. Control movers: 10% sample of workers who move between non-cartel BEC firms in 2011–2014. Occupations from CBO 2002 major groups. Source: script 32.

The profile of cartel movers is clear—and it is not the profile of knowledge carriers. Managers and professionals are *under*represented among movers (9.9%) relative to stayers (12.9%) and control movers (12.3%). The modal cartel mover is a service or sales worker (55% of movers vs. 37% of stayers), female (60%), earning 2.4 minimum wages, with lower educational attainment than stayers. This is a displacement profile, not a knowledge-transfer profile. The workers who carry pricing strategies and coordination know-how—directors, professionals, long-tenured specialists—are disproportionately the ones who *stay*.

The finding has a direct implication for the collusion-contagion concern. If cartel firms were shedding coordination-capable workers to competitors, we would expect high-skill, high-wage movers. Instead, the outflow is dominated by operational workers whose departure is consistent with generic firm contraction rather than strategic knowledge diffusion.

6.9. Is the pattern cartel-specific? An exit placebo

The temporal decomposition and event study show that cartel firms' outflows increase after conduct ends. But is this a cartel-specific phenomenon, or would any firm that contracts in thin markets generate the same pattern? To test this, we construct a placebo from 3,335 non-cartel BEC firms that exit the platform (last bid between 2011 and 2014, at least two active years), and run the same event-study specification as in Section 6.6, with the exit year replacing the conduct-end year.

Table 8: Event Study: Cartel Firms vs. Non-Cartel Exit Firms

τ	$\hat{\beta}$ (exit)	SE	$\hat{\beta}$ (cartel)	Sig. (exit)
-3	$+0.005$	0.003	-0.003	$*$
-2	-0.001	0.002	$+0.014$	
-1	(reference)		(reference)	
0	$+0.015$	0.002	$+0.025$	$***$
$+1$	$+0.021$	0.003	$+0.035$	$***$
$+2$	$+0.018$	0.003	-0.001	$***$
$+3$	$+0.015$	0.003	$+0.019$	$***$
$+4$	$+0.009$	0.003	$+0.040$	$***$
Mean $\tau \geq 0$	$+0.016$		$+0.019$	

Notes: Exit firms: non-cartel BEC firms with last bid in 2011–2014 and ≥ 2 active years ($N = 3{,}335$). Cartel coefficients from Section 6.6 ($N = 41$). Firm and year FEs; firm-clustered SEs. Source: scripts 28 and 33.

The exit-placebo coefficients are highly significant ($t = 5$–8, reflecting the much larger treated sample) and economically similar to the cartel coefficients: post-event outflows average +1.6 pp for exit firms versus +1.9 pp for cartel firms, a ratio of 1.25. The patterns are not statistically distinguishable.

This is a sobering finding. The post-conduct outflow pattern that drives the full-panel worker-flow AUC is not cartel-specific—it is a generic consequence of firm contraction in thin labor markets. Any firm that stops bidding on BEC generates elevated worker outflows to competitors in subsequent years, at roughly the same magnitude as cartel

firms after their conduct period ends. Cartel enforcement is *one cause* of firm contraction, but the labor-market response it triggers is not distinctive.

The result sharpens the paper's contribution in two ways. First, it reinforces the look-ahead bias warning: full-panel network screens absorb post-event labor reallocation regardless of whether the event is a cartel conviction, a voluntary exit, or any other firm contraction. The bias is *generic*, not cartel-specific, which makes the methodological caveat broader and more important. Second, it disciplines the interpretation: the matched employer–employee record captures the labor-market aftermath of firm exit in thin markets, not a cartel-specific enforcement effect.

6.10. Reinterpreting the thickness gradient

The temporal conditioning results change the interpretation of the market-thickness analysis, but not its empirical content. We split the benchmark by median bidders per item code (computed on clean controls). Table 9 and Figure 2 report AUCs by thickness group.

Table 9: Screen Performance by Market Thickness

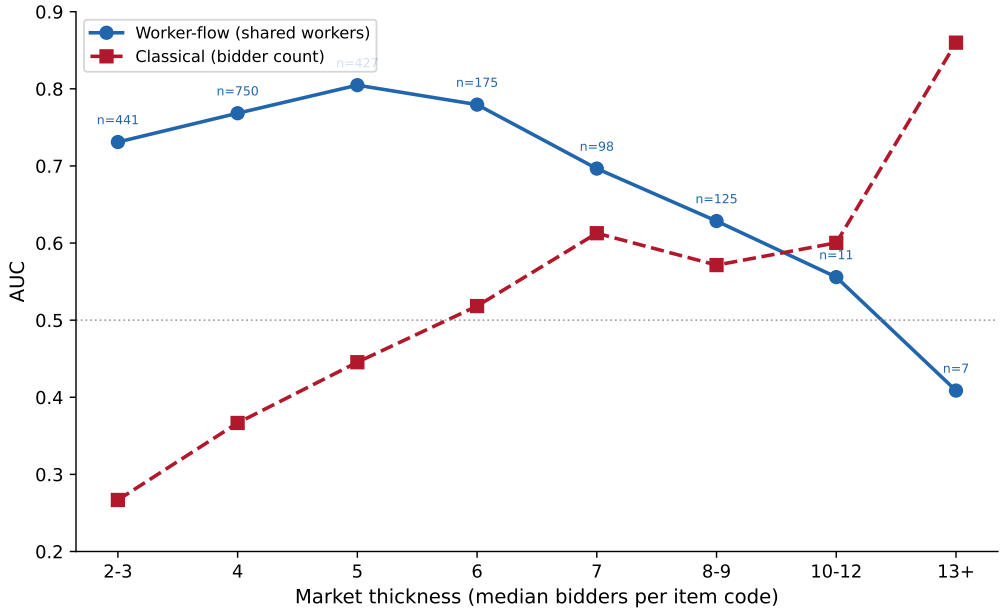
	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
Thin (≤ 5 firms)	1,618	217,662	0.447	0.436	0.400	**0.774**	**0.758**
Thick (> 5 firms)	416	224,074	0.594	0.604	0.604	0.699	0.694
All (benchmark)	2,319	527,259	0.523	0.531	0.555	**0.750**	**0.739**

Notes: Market thickness is the median number of bidders per auction for each item code, computed on clean-control auctions with ≥ 5 observations. "Thin" = median ≤ 5; "Thick" = median > 5. Classical screen AUCs use "lower is cartel" convention. Worker-flow AUCs use full-panel shared workers. 285 cartel-active auctions with item codes having < 5 clean observations are unclassified and excluded from thin/thick rows but included in "All." Source: script 24.

The gradient for *classical screens* is robust and does not depend on the temporal-conditioning critique: in thin markets, classical screens fire in the wrong direction (AUC 0.40–0.45); in thick markets, they recover to 0.60. This pattern—which uses contemporaneous bid data, not panel employment data—is a clean empirical finding about where traditional detection methods fail.

The gradient for *worker-flow screens* requires reinterpretation. The full-panel worker-flow AUC is higher in thin markets (0.77) than in thick ones (0.70). In light of the temporal decomposition, this gradient tells us not where *detection* is easiest, but where *post-conduct labor reallocation* is most concentrated. Thin markets, by definition, have

Figure 2: Screen Performance by Market Thickness



Notes: Each point represents the AUC within a thickness bin. Worker-flow AUC uses shared workers (full panel); classical AUC uses bidder count (negated). Sample sizes (n) refer to cartel-active auctions per bin.

few employers capable of absorbing displaced workers—so post-conduct flows between cartel firms and rivals are mechanically denser, producing higher retrospective AUCs.

6.11. Reinterpreting the specialization placebo

The specialization-placebo decomposition (Table 10) similarly retains its empirical content but shifts in interpretation.

The original interpretation was that 44% of the worker-flow signal reflects natural thin-market labor circulation and 56% reflects cartel-specific coordination. In light of the temporal evidence, the "cartel-specific residual" is more precisely a *post-conduct displacement residual*: cartel firms shed more workers to rivals than same-market non-cartel firms do, because enforcement-driven contraction is concentrated in cartel firms. The specialization component remains what it was—thin markets generate spontaneous worker overlap among all firms—but the cartel residual is not evidence of coordination during the cartel period. It is evidence of accelerated labor reallocation after the cartel period ends.

Table 10: Specialization-Placebo Decomposition of Worker-Flow Signal

Sample	N	% with ≥ 1 shared worker	Mean Jaccard	AUC vs. clean
Cartel-active	2,321	57.9%	0.0022	0.740
Placebo (same market)	5,313	31.1%	0.0012	0.605
Clean (global)	523,414	9.9%	0.0008	—
Signal decomposition (Jaccard AUC relative to clean baseline)				
Total signal (cartel vs. clean − 0.5)				+0.240
Specialization component (placebo vs. clean)				+0.105 (44%)
Cartel-specific residual (cartel vs. placebo)				+0.138 (56%)

Notes: Pregão, $|MV| \leq 0.10$. Placebo consists of non-cartel firm pairs bidding on the same BEC item code in the same year. AUC computed via Mann–Whitney U (Jaccard similarity, higher is cartel). Source: v3 specialization placebo.

6.12. Sample concentration and cartel-by-cartel performance

The cartel-active benchmark is not equally distributed across the seven convicted cartels. Table 11 reports the composition and per-cartel AUCs.

Table 11: Per-Cartel Sample Sizes and AUCs

Cartel	N	% of sample	CV AUC	SW AUC (full)
Medicamentos	1,977	85.3	0.501	0.780
Sacos de lixo	224	9.7	0.633	0.530
Trens e metrôs	85	3.7	0.680	0.592
Merenda escolar	26	1.1	0.820	0.941
Transporte escolar	6	0.3	0.191	0.527
Cafeteria aeroporto	1	0.0	—	—
Aquecedores solares	0	0.0	—	—
All	2,319	100	0.523	0.750

Notes: Pregão, $|MV| \leq 0.10$. CV AUC uses "lower is cartel" convention. SW AUC uses full-panel shared workers. Aquecedores solares has zero cartel-active auctions in the close-margin benchmark. Source: script 25.

The medicamentos cartel accounts for 85% of the treated sample. Its full-panel worker-flow AUC (0.780) essentially *is* the headline result. Meanwhile, *sacos de lixo* and *trens e metrôs* show classical screen AUCs above 0.63 but worker-flow AUCs near chance (0.53–0.59), consistent with the thickness gradient: these are thicker markets where classical screens work and where post-conduct labor reallocation is less concentrated. The concentration is a caveat for all results in this paper. It does not invalidate the temporal decomposition—the post-conduct pattern is present in the medicamentos case

and absent in pre-conduct data—but it limits external validity. Replication on larger ground-truth sets is a priority.

6.13. Robustness: permutation falsification

We randomly reassign the cartel-active label 1,000 times and recompute the full-panel AUC. The permuted AUCs cluster at 0.500 (sd = 0.003, 95% interval $[0.494, 0.505]$). The real AUC of 0.75 lies at the 100th percentile ($p < 0.001$): the full-panel signal is non-random. But as the temporal decomposition shows, it is retrospective rather than prospective.

6.14. Model predictions vs. empirical findings

Table 12 summarizes the correspondence between the model's predictions (Section 4.7) and the results, including temporal conditioning.

The classical-screen predictions (P1 for classical screens) are cleanly confirmed and robust. The worker-flow predictions require reinterpretation: the model posits that worker flows carry signal because they reflect coordination; the temporal evidence shows they carry signal because they reflect post-conduct displacement. The institutional-thinness mechanism is still operative—thin markets are where post-conduct reallocation concentrates—but the channel runs through enforcement aftermath rather than active coordination.

6.15. Selection and the interpretation of the benchmark

The temporal conditioning results largely supersede the selection concern. The full-panel AUC is inflated not by selection but by look-ahead. Once temporal conditioning is applied, the AUC falls to 0.52 regardless of selection on detectability.

For the record, none of the seven CADE cases lists employer–employee data or worker flows as part of the evidentiary basis (Table 13). The selection concern remains relevant for interpreting the post-conduct displacement pattern: if CADE preferentially convicts cartels in markets where labor reallocation is naturally high, the post-conduct AUC could partly reflect market selection rather than enforcement effects.

Table 12: Model Predictions and Empirical Results

Prediction	Empirical test	Result
P1. In thin markets, classical AUCs ≈ 0.5; worker-flow AUCs > 0.5.	Table 9: median split	Classical AUC = 0.40–0.45 (inverted). Full-panel worker-flow AUC = 0.77, but $\approx$ 0.52 after temporal conditioning. **Classical gradient confirmed; worker-flow gradient is retrospective.**
P2. Same-market placebo shows elevated worker-flow; cartel pairs show more.	Table 10: three-way decomposition	Both components present, but cartel residual driven by post-conduct flows. **Pattern confirmed; interpretation revised.**
P3. Worker-flow signal symmetric in winner/loser role.	Section 6.3: role decomposition	AUC = 0.74–0.76. **Confirmed.** Consistent with both coordination and displacement.

Table 13: Evidence Channels in the Seven Convicted Cartels

Cartel (sector)	CADE process	Period	Primary evidence channel
Merenda escolar	08012.010022/ 2008-16	2006–2013	Procurement documents (40,000+ docs analyzed); geographic market division
Trens e metrôs	08700.004617/ 2013-41	1999–2013	Leniency and investigation; subcontracting arrangements between winners and losers
Aquecedores solares	08012.001273/ 2010-24	2009–2010	Indirect evidence only (bid-pattern analysis); no direct evidence of agreement
Transporte escolar	08700.005876/ 2019-85	2019	IP address coincidence during electronic bidding
Cafeterias Infraero	08700.007278/ 2015-17	2014	Investigation of in-person tender procedures
Sacos de lixo	08700.005789/ 2015-02	2008–2014	Police operation (*Op. Colludium*, MP Bauru); dawn raids and document seizures
Medicamentos	08012.002222/ 2011-09	2007–2011	Hub-and-spoke investigation; leniency and administrative proceedings

Notes: Evidence channels documented from CADE public press releases and Tribunal decisions. None lists matched employer–employee data, RAIS records, or inter-firm worker flows as part of the evidentiary basis.

Table 14: Unified Benchmark: All Screens on the Same Close-Margin Sample

	Pregão ($N_t = 2{,}319$)		Convite ($N_t = 263$)	
Screen	Cohen's d	AUC	Cohen's d	AUC
Panel A: Classical bid-distribution screens				
Coefficient of variation (CV)	−0.07	0.523	−0.01	0.515
Bid spread (S)	−0.10	0.531	+0.04	0.502
Margin of victory (MV)	+0.17	0.457	+0.14	0.456
Number of bidders (N)	−0.24	0.555	+0.06	0.467
Panel B: Network-based screens				
Shared workers (W_{ij})	+0.06	**0.750**	+0.47	**0.704**
Jaccard similarity (J_{ij})	+0.19	**0.739**	+0.09	**0.701**
Co-bidding density (C_{ij})	+0.14	0.695	−0.04	0.525

Notes: All screens evaluated on the unified close-margin benchmark ($|MV| \leq 0.10$) against a common clean control ($N_0 = 527{,}259$ pregão; $N_0 = 519{,}557$ convite). Classical screen AUCs use the "lower is cartel" convention; values above 0.5 indicate detection in the predicted direction. Network screen AUCs use "higher is cartel." Bold indicates AUC ≥ 0.70. Cohen's d is the standardized mean difference (pooled SD). Source: script 23.

7. Discussion and Conclusion

This paper set out to test whether worker-flow screens could detect procurement cartels where classical screens fail. The answer is no—but the reason why is more interesting than a simple null.

We assembled a conviction-anchored benchmark linking seven CADE-convicted procurement cartels to 2.6 million competitive auctions and the universe of formal employment in São Paulo. Four findings emerge.

The first is a **positive detection result**. Classical screens fail in the aggregate (AUC 0.46–0.56), but this null is a composition effect. Disaggregating by coordination technology reveals that classical screens detect bid rigging at AUCs of 0.65—operationally useful for triage. They are blind to the hub-and-spoke cartel that dominates the sample. The coordination technology determines which screen works: authorities should match the screen to the suspected mechanism.

The second is a **warning about network-based screens**. Worker-flow screens appear to fill the gap left by classical screens, with an AUC of 0.75—but this result does not survive temporal conditioning. When worker flows are restricted to information available before each auction, the AUC falls to 0.50. Co-bidding screens show the same pattern (0.70 → 0.55). Neither network screen achieves prospective detection power on

this benchmark.

The third finding is a **methodological caveat** with broad applicability. The gap between the full-panel AUC (0.75) and the temporally conditioned AUC (0.50) documents a form of look-ahead bias that affects any detection benchmark built from administrative panel data. When a network screen is evaluated against conviction-anchored ground truth, post-enforcement changes in the network—labor reallocation, ownership restructuring, board turnover—mechanically inflate the apparent detection power. This bias is not specific to worker-flow screens: it applies equally to ownership networks (Asai and Charoenwong, 2026), board-interlock measures (Herrera-Caicedo et al., 2025), and any network layer constructed from longitudinal administrative records. The remedy is straightforward—condition the network measure on temporal precedence—but the bias has not, to our knowledge, been documented or tested in the detection literature.

The fourth finding is **substantive**. The temporal decomposition reveals that worker flows between cartel firms and their rivals are concentrated *after* cartel conduct periods end, not before or during. The post-conduct component alone produces an AUC of 0.77; the pre-conduct component produces exactly 0.50. The flows are directional—workers move predominantly from convicted firms to competitors—and the ratio intensifies from 1.4:1 in 2010 to 4:1 by 2016–2017. An exit placebo shows that the pattern is not cartel-specific: non-cartel firms that stop bidding on BEC generate outflow spikes of similar magnitude (+1.6 pp vs. +1.9 pp for cartel firms). The labor-market aftermath is a generic consequence of firm contraction in thin markets, and cartel enforcement is one cause of such contraction. Characterizing the movers confirms this interpretation: workers who leave cartel firms are predominantly operational staff in service and sales roles, not the managers and professionals who might carry coordination know-how.

Policy implications. For competition authorities building data-driven screening pipelines, the message is direct: network-based screens evaluated on historical benchmarks may overstate their real-time usefulness if the evaluation does not condition on temporal precedence. The bias is generic—it arises whenever firms contract and workers reallocate, regardless of whether the contraction is cartel-related—which makes it a concern for any panel-based detection benchmark, not just ours. Authorities adopting worker-flow or ownership-network screens should validate them prospectively, not retrospectively. The exit-placebo result also mitigates one policy concern: because the workers who leave cartel firms are predominantly operational staff rather than managers or coordination-

capable professionals, the risk that enforcement-driven labor reallocation diffuses collusive know-how to competitors appears low.

Limitations. The principal caveat remains the small number of convicted cartels: seven cases, one of which (medicamentos) accounts for 85% of the cartel-active sample. The temporal decomposition results are driven disproportionately by this case, and replication on larger or independent ground-truth sets is a priority. The RAIS panel ends in 2017 while CADE judgments for most cartels arrive after 2019, so the "post-conduct" window we measure captures the investigation and pre-judgment period, not the post-judgment period. A definitive test of enforcement-driven reallocation would require RAIS data extending through the judgment dates. Finally, the benchmark captures only cartels that were detected and convicted; the worker-flow patterns of undetected cartels remain unknown.

Looking ahead. The broader lesson is about what administrative records actually capture. Employment data do not record collusion in progress—the personnel flows that sustain coordination, if they exist, are too subtle or informal to leave traces in annual employment censuses. What employment records *do* capture is the aftermath: the reallocation of human capital that follows enforcement. Following the workers, rather than the bids, reveals not the cartel itself but the institutional disruption it leaves behind. Understanding that disruption—its magnitude, its direction, and its consequences for market competition—is the next step.

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A. Appendix

A.1. Bandwidth sensitivity

Table A.1 reports AUCs for all screens across six bandwidth choices for the close-margin envelope. The benchmark bandwidth of $|MV| \leq 0.10$ used throughout the paper is neither the most nor the least favorable for the worker-flow screens. At narrower bandwidths (0.02, 0.05), worker-flow AUCs are marginally higher (0.76, 0.75) but sample sizes are smaller; at wider bandwidths (0.20, 0.50), AUCs decline modestly (0.74, 0.72) as the sample dilutes toward the full population. Classical screen AUCs follow the same monotonic pattern. The gap between the best classical screen (number of bidders) and the worker-flow screens remains stable at approximately 18–20 percentage points across all bandwidths.

Table A.1: AUC Sensitivity to Close-Margin Bandwidth (Pregão, Cartel-Active)

$\|MV\| <$	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
0.02	989	277,038	0.546	0.558	0.580	0.758	0.747
0.05	1,700	416,582	0.539	0.548	0.565	0.760	0.749
0.10	**2,319**	**527,259**	**0.523**	**0.531**	**0.555**	**0.750**	**0.739**
0.15	2,607	573,196	0.518	0.526	0.554	0.745	0.734
0.20	2,784	602,883	0.517	0.525	0.555	0.743	0.732
0.50	3,277	693,331	0.518	0.526	0.552	0.735	0.723

Notes: Each row reports AUCs from the unified benchmark (script 23) at the indicated bandwidth. Classical screen AUCs use the "lower is cartel" convention (negated scores). Bold row indicates the benchmark bandwidth used in the main analysis. All AUCs computed via Mann–Whitney U with average-rank tie-handling.